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Research paper

Democratizing eye-tracking? Appearance-based gaze estimation with improved attention branch

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ABSTRACT

Appearance-based gaze estimation in 2-dimensional screen coordinates–the prediction of the users' gaze from webcam footage–cannot yet compete in accuracy with infrared (IR) eye trackers. Yet by circumventing the constraints of requiring dedicated hardware, it shows great potential in many technological industries, as evidenced by some readily available commercial solutions, bringing democratization of eye tracking closer to the people. We present Residual Appearance-based Gaze Estimation network (RAGE-net), a novel convolutional neural network for gaze estimation without need of calibration, utilizing a fraction of computational resources required by similar networks, while also achieving competitive accuracy. The angular error is measured as 4.08° in the MPIIFaceGaze dataset (Max Planck Institute for Informatics Faze Gaze) and 3.96° in the MPIIGaze dataset. The architecture's principles, covered by a comprehensive ablation study, include an attention branch, residual learning, weight sharing between eye channels, batch normalization and an eye image input normalization pipeline that removes dependence on full face input. With RAGE-net, we conduct an applicability study for gaze estimation approaches of similar accuracy for interpreting on-screen gaze in praxis. Findings demonstrate low heatmap validity, with coarse heatmaps as potential adaptation to approximate IR eye tracking. The effects of environmental factors such as camera position, illumination, distance and glasses are analyzed in-depth.

1. Introduction

Knowing the focus of a user's visual attention has invaluable uses in a spectrum of fields, including human–computer interaction, psychology, medicine, software engineering, education, and ecommerce (Bhatnagar and Orquin, 2022; Bâce et al., 2020; Suslow et al., 2020; Fei et al., 2021; Motoki et al., 2021; Cho et al., 2019; Liu et al., 2021b; Wang et al., 2020; Klaib et al., 2021; Wang et al., 2022). Gaze tracking (extrapolating eye movements to tell where a person is looking) is the typical source of information on visual attention, followed by less direct alternatives such as movements of the mouse (Arapakis and Leiva, 2020). The reason for the existence of alternatives is that while modern gaze tracking is highly accurate (0.5–1° visual angle error) (Sharafi et al., 2020), it requires specialized–often expensive–software and hardware (a designated infrared camera). This renders pursuits to democratize gaze tracking, spreading its availability to anyone, at scale and outside of controlled environments, impractical at best.

Unlike eye trackers, webcams are built into almost every laptop and phone. External webcams are common household devices, especially in the post-COVID19 world. This positions webcams as a cheap and widespread source of data for methods that try to estimate the direction of a user's gaze based on the appearance. The sphere of appearance-based gaze estimation in 2D screen coordinates has been the subject of research in the recent years (Huang et al., 2023; Cheng and Lu, 2022; Murthy and Biswas, 2021; Zhou et al., 2019; Yan and Tang, 2021; Guo et al., 2021; Liu et al., 2021a). Not competing directly with infrared gaze tracking in terms of accuracy, it can nevertheless be considered a serviceable substitute in use cases limited by the end-user's natural environment. Commercial webcam-driven gaze tracking solutions are readily available (e.g., by Tobii[1]) or are the subject of research and development by user experience and usability research tool providers (e.g., by UXtweak[2]). When referring to gaze estimation in this paper, we refer to the category of screen and webcam-oriented methods described above. In literature, gaze estimation also encapsulates a broader range of approaches and problems (Pathirana et al.,

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[1] Tobii eye tracking: https://www.tobii.com/.
[2] UXtweak user experience research platform: https://www.uxtweak.com/.

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2022), such as with multiple subjects, in image coordinate systems or subject/camera coordinate systems, which are outside the scope of study of this research.

We propose a new evolution of webcam-driven real-time gaze tracking via a convolutional neural network (CNN). Our deep learning network architecture utilizes the attention principle (Murthy and Biswas, 2021) with batch normalization, an attention branch focusing the network's own attention on sections of eye images relevant to assessing where the user's gaze is pointed. With the angular error of 4.08° in the MPIIFaceGaze dataset and 3.96° in the MPIIGaze dataset, performance-wise it is comparable to more complex similar models (Murthy and Biswas, 2021; Ren et al., 2023), but with a leaner design that necessitates significantly fewer parameters and floating point operations (FLOPs).

Besides proposing our novel model, we comprehensively analyze its applicability (and by extension, that of similarly accurate models). The error metric typically reported with gaze estimation models in literature is satisfactory for comparison purposes, but it cannot be used to directly interpret the gaze-estimation model's practical utility. Although datasets from realistic usage conditions are commonly used to train CNN gaze estimation models, there is a lack of understanding of how accuracy is impacted by specific factors of the environment encountered in real-use scenarios, such as glasses worn by the user, suboptimal lighting conditions or the webcam being positioned differently. Zhang et al. (2019) have analyzed the effect of some of these factors on the accuracy of appearance-based gaze estimation for desktop computer screens in comparison with the Tobii EyeX eye tracker. They demonstrated that wearing glasses or being outdoors can cause higher estimation errors. We broaden the understanding of the patterns in which the errors manifest to inform methods for mitigating the environment's effects in gaze estimation and its interpretation. We make the following contributions in this regard:

- expanded exploration of additional environmental factors (levels of illumination, camera placement),
- qualitatively-oriented analysis and visualization of the impacts of individual factors,
- comparison with the TX-300, an eye tracker more representative of the high accuracy of IR eye tracking (~0.5°) compared to the Tobii EyeX (~1°), and
- analysis of the applicability and reliability of appearance-based gaze visualizations in a manner for which highly-accurate but less-accessible infrared gaze tracking is used.

To outline the structure of this paper, Section 2 highlights the key contributions, Section 3 summarizes existing research in the field of webcam-driven gaze tracking and Section 4 specifies our proposed network. The ablation study that underpins the data-driven design proposal, along with the experiments subjecting our network to analysis of its applicability are described in Section 5. Results are discussed in Section 6. Section 7 addresses their limitations and potential for future exploration. Section 8 delivers the conclusion.

2. Research objectives and contributions

Our goal in this work is to enhance and examine the efficiency and effectiveness of 2D appearance-based gaze estimation, primarily in desktop computer environments in realistic conditions. For this purpose, we conduct an introspective experimental investigation of the practical utility of appearance-based gaze estimation, involving a neural network architecture designed as an accurate representative model that leverages current deep learning approaches. The principal contributions of the research covered by this paper are:

- Proposal of RAGE-net (Residual Attention-based Gaze Estimation network), a novel lightweight convolutional neural network architecture for appearance-based gaze estimation without calibration that performs comparably to existing related approaches while decreasing the network size and computational complexity.
- Comprehensive ablation study with the dataset MPIIFaceGaze to empirically support the design decisions implemented in RAGE-net
- Assessment of the accuracy of heatmaps calculated from gaze paths estimated via RAGE-net.
- Examination of the comparative impact of individual environmental factors altering the eyes' appearance (illumination, distance from the screen, glasses, position of the camera) on the error of RAGE-net gaze estimation and on feature-based (infrared) gaze tracking.
- Discussion of the limitations and approaches for interpretation of visual attention involving appearance-based gaze estimation.

3. Background and literature review

Seeking to evaluate the usefulness of appearance-based gaze estimation with regular commercial cameras for eye tracking studies, we present an overview of key terms and concepts from the domain of eye tracking. Commonly used techniques for measuring visual attention in stimuli are summarized. The review concludes by covering convolutional neural networks and their application for gaze estimation.

3.1. Human visual perception

The human visual system consists of the eyes, the optical nerve and the visual cortex of the brain. The fovea is the most photosensitive point on the eye's retina (Leigh and Zee, 2015). To assess the accuracy of gaze tracking in precise and constrained 2D environments, such as on a computer screen, it is salient to understand that as a characteristic of foveal vision, at any given moment, the human eye can sharply observe only a narrow 0.5° angle of the visual field.

To change the object of observation, the gaze needs to shift constantly (Ogmen and Herzog, 2010), allowing for perception of a continuous scene. A scanpath (Castner et al., 2020) is trajectory of eye movement behavior, a vector of coordinates within the environment as they were visually scanned. Typical eye movements are characterized by two types of behavioral patterns — fixations and saccades (Duchowski, 2017). Fixations are moments when the gaze position is kept stationary, allowing for cognitive processing of a static observed area. Fixations endure in place for 150–600 ms before the gaze point moves again. Ninety percent (90%) of all gazing time is spent on fixations. High-speed eye movements between fixations, typically 10–100 ms long, are called saccades (Squire et al., 2014).

A plurality of visualizations have been developed to facilitate human comprehension and exploration of gaze behavior. Comparing visualized gaze estimations to ground truth visualizations obtained via baseline infrared gaze tracking methods yields itself as a practical indicator of the reliability with which these estimations can be interpreted. In broad terms, there are two classes of gaze tracking visualizations — point-based and based on areas of interest (Blascheck et al., 2017). Point-based visualizations such as attention heatmaps, scanpaths and gaze timelines deliver spatial and temporal information on position of gaze and fixations. Areas of interest (AOIs) incorporate the additional context of the object that the gaze was being fixated on. AOI visualizations include AOI timelines and relational tables. Eye tracking visualizations can be further taxonomized according to other criteria, such as:

- two-dimensional (2D) vs. three-dimensional (3D),
- spatial vs. temporal vs. spatiotemporal,
- static vs. animated,
- interactive vs. noninteractive.

3.2. Gaze tracking and related techniques

Design and discussion of the proposed gaze estimation model and the method providing the ground truth merit a disambiguation of the terms "eye tracking", "gaze tracking" or "gaze estimation". These terms are commonly used semi-interchangeably to refer to approaches, techniques and technologies for recognizing where a person is looking (Klaib et al., 2021; Liu et al., 2020). However, there are key nuances in the aspects that they represent (Khan and Lee, 2019). The first step – harnessing data on eyes, their presence, shape and behavior – can be called eye tracking. The second step – utilizing the eye tracking data to detect the point of gaze – can be called gaze estimation. Together, they can be referred to as gaze tracking.

In some gaze tracking approaches, calibration is a common step to accommodate individual users and the conditions of their environment. Models that take the individual's characteristics into consideration are known to perform better than generalistic models (Li et al., 2020). Standard calibration involves tasking the participant with shifting their gaze between a k-number of fixed positions on a screen. The disadvantage of reliance on calibration in application contexts is that it presents a barrier for utilizing gaze tracking. Even the simplest calibration intrusively interrupts the user's normal activity, presenting an issue for usability (Zhang et al., 2019). Acquisition of data in the users' natural environment should preclude such distractions, to prevent affecting the observation or frustrating the subject. Therefore, the focus on this work is on calibrationless gaze estimation.

3.2.1. Eye tracking

Among eye tracking methods that provide inputs for gaze estimation, there is typically an inverse relationship between their accuracy and how obtrusive and cumbersome they are (Sigut and Sidha, 2011). Remote infrared (IR) oculography is popular for its limited obtrusiveness and reasonably high accuracy and precision (0.5–1° and 0.01–1° respectively) (Sharafi et al., 2020). Infrared eye-trackers emit infrared light invisible to humans, then detect its reflection off the cornea. They are undisruptive since they can be placed in the environment, yet they are not completely barrier-free, since they require dedicated equipment to operate. Eye tracking hardware and software developed by Tobii[3] is commonly used in research (Nyström et al., 2021; Niehorster et al., 2020a,b). Due to the high price point and difficulty of obtaining special eye tracking equipment, availability of infrared eye tracking is hampered in practice by its low penetration among the populace outside of dedicated eye tracking labs.

Video oculography encompasses a growing body of approaches that utilize cameras (infrared or visible spectrum). Methods exist utilizing head-mounted cameras (Meyer et al., 2022; Khan et al., 2018), stereo cameras (Hausamann et al., 2020; Bulling and Gellersen, 2010) as well as single cameras (Hosp et al., 2020). We delve into methods for the more commonplace visible-spectrum single cameras (regular webcams) as part of the endeavor to dissolve the technological boundaries imposed on gaze tracking.

3.2.2. Gaze estimation

In gaze estimation, three paradigms command attention — feature-based, model-based, and appearance-based (Hansen and Ji, 2010). Feature-based methods rely on the exploitation of specific physical properties of the eyes to gauge the direction where the eyes are looking. The popular pupil center corneal reflection technique (Chugh et al., 2021) used in commercial infrared gaze tracking is the prime example. Model-based approaches construct a 3D model of the eyes. Appearance-based gaze estimation (Zhang et al., 2019) typically involves machine learning to regress the angle and position of gaze based on the image data. A significant benefit of this approach is that it can be applied even in conditions bound by the limitations of an off-the-shelf webcam (Jiang et al., 2019).

[3] Eye-tracking tech company Tobii: https://www.tobii.com/.

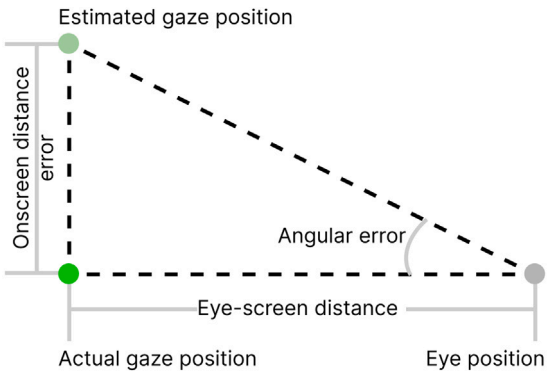


Fig. 1. Angular error in gaze estimation is calculated as a trigonometric function of the distance between the estimated and real gaze point on the screen, and the distance of the eyes from the screen. Higher angle means less accurate regression.

Appearance-based gaze estimation is a data-based approach. It requires a notable amount of image data mapped to gaze positions for training of a prediction model. Images are dissected into regions of interest, extracted as high-dimensional vectors for the regression task of predicting gaze point coordinates (Huang et al., 2017b). Utilization of image vectors renders appearance-based techniques inherently sensitive to aspects of the scene such as lighting conditions, the individual's facial features, and the angle of the face and camera (Khan and Lee, 2019; Zhang et al., 2015). The training sample should thus account for various possible conditions.

Machine learning methods that have been employed for appearance-based gaze estimation include random forest (Kacete et al., 2017; Huang et al., 2017b; Wang et al., 2016), k-nearest neighbors (KNN) (Wang et al., 2018; Wood et al., 2016), Support Vector Machine (SVM) (Melesse et al., 2020; Wood et al., 2016; Wu et al., 2014). In comparison to the above methods, convolutional neural networks (CNNs) accomplish the best performance (Zhang et al., 2015) and as such are the focus of this work.

3.3. Measuring gaze estimation

Angular error (see Fig. 1) is a metric used to gauge the accuracy of gaze estimation for comparison between models. Expressed in degrees (°), it is sometimes also measured in "dva" (degrees of visual angle). Unlike pixel or physical distance oriented metrics, angular error is decoupled from any specific screen size or distance of eyes from the screen. It is calculated (Li et al., 2018) as:

$$angular_error = arctan\left(\frac{onscreen_distance_error}{eye_screen_distance}\right) \quad (1)$$

Onscreen distance error is the physical Euclidean distance of the estimated and the expected gaze point (the actual position of gaze on the screen). Eye-screen distance is the distance of the center of the individual eyes from the expected gaze point.

3.4. Convolutional neural networks in gaze estimation

Convolutional neural networks (CNN) are a type of artificial neural networks that have been exploding in popularity during the last decade, with successful applications in the fields of computer vision and natural language and speech processing (Li et al., 2022; Gu et al., 2018; Punyani et al., 2020). As a deep learning method, they are capable of learning both low and high-level patterns of spatial hierarchy. The inputs they operate with are data matrices (tensors). In 2D images,

a tensor is a grid with h-number of rows, w-number of columns and possibly multiple channels (e.g., three values corresponding to the colors of the RGB color model).

Owing to the boom of CNNs, an abundance of different architectures has been proposed, striving to improve performance, efficiency but also adaptability and transportability of CNNs for solution of image and sound-related machine learning problems. Khan et al. (2020) divide architectural directions into seven classes — spatial exploitation, feature-map exploitation, channel exploitation, multipath, depth, width, and attention. Typical aspects differentiating architectures are the topology of processing units and design of blocks — the cascade of convolution and pooling layers (Cong and Zhou, 2023). We focus on VGGNet and ResNet, adaptable architectures with key beneficial characteristics for which they are considered as modules of RAGE-net.

3.4.1. VGGNet

VGGNet (Jun et al., 2018; Simonyan and Zisserman, 2014) is an architecture that uses small 3 × 3-dimension kernel convolutions (as opposed to single 5 × 5 or 7 × 7), the minimum size for horizontal and vertical convolution. 5 subsequent blocks consist of convolutions and max pooling (kernel size 2 × 2 and stride 2). Predictive ability is improved by decreasing the number of weights and increasing the number of activation functions. VGGNet shows that network depth plays a beneficial role in task-solving accuracy, with the deepest variants (16 and 19) being the best at classification. VGGNet can be considered a reasonable baseline for the gaze estimation task, given its success demonstrated in related works (Murthy and Biswas, 2021; Chen and Shi, 2019). Its depth and simple convolution allows for more complex pattern processing over the likes of AlexNet.

3.4.2. ResNet

ResNet (He et al., 2016) addresses the problem of network degradation, a phenomenon where adding layers to improve the accuracy of a neural network only works up to a certain point, after which adding more layers actually leads to an increased error on the training set. This is distinct from overfitting, where the error appears in the testing sample. ResNet's solution hinges on placing shortcut connections between more distant layers that skip the blocks between them. Instead of learning the intended mapping, such skipped blocks instead learn residual mapping. Promotion of longer learning without degradation may be the reason for ResNet's growing popularity in the problem area of gaze estimation (Pathirana et al., 2022). When focusing on performance and the potential for real-time application, ResNet can also be preferred over DenseNet for its lower computational complexity (Huang et al., 2017a).

3.4.3. State-of-the-art approaches

For calibrationless gaze estimation in constrained environments, CNNs have been the center of attention owing to their ability to reliably outperform alternative approaches. (Zhang et al., 2015) use the SURF method (an alternative to SIFT that trades off some of its robustness for working in real time (Kuric and Bielikova, 2015)) for facial detection, with a generic head shape model calculated as a mean of all participants' head shape models obtained from data captured with a stereo camera. The image input is scaled and rotated to normalize its 3D pose, bringing the face to an equal distance and making the face's and the camera's x axes parallel to one another. The input for the CNN is composed of cropped eye images and the head angle vector. The LeNet architecture and a linear regression layer are used to predict the gaze angle in the camera's coordinate system. Compared to other machine learning models—random forest, adaptive linear regression, k-nearest neighbor and support vector regression—this approach achieves the best result: an angular error of 6.3°. The same authors later improved the angular error to 5.4° by utilizing the 16-layer VGGNet (Zhang et al., 2017b).

Krafka et al. (2016) propose a multi-channel architecture iTracker that has cropped images of both eyes and the face image serving as its inputs, also replacing the rotation vector with a face grid—a binary mask that shows the position of the face in the shot. Zhang et al. (2017a) achieve an improvement to the error of 4.8° while using only full face images. Spatial weights are used under the assumption that some areas besides the eyes are greater or lesser determinators of the gaze angle, while other areas such as the background do not help determine the gaze direction at all.

More recent innovations for appearance-based gaze estimation include dilated convolutions (Chen and Shi, 2019; Murthy et al., 2021), bi-directional long short-term memory (LSTM) (Zhou et al., 2019), accounting for asymmetry between the two eyes (Cheng et al., 2020b), coarse-to-fine-grained network strategies (Cheng et al., 2020a; Hu et al., 2023), and use of a transformer encoder (Cheng and Lu, 2022). See Table 1 for a discriminative summary of related works and their results juxtaposed with ours. Note that a survey of the works included in the table shows that the nature of the distance—whether it is actual or normalized—is sometimes left undisclosed by the original research, creating potential inconsistency between the formulas used for error in different works. To report the angular error of the proposed network, normalized eye-screen distance of 600 mm is used, to correspond with the pose normalization, which maps the face to a 3D model distanced 600 mm from the screen — see 4 Proposed architecture.

The attention mechanism comprises the assignment of weights to specific areas of the input dynamically, emphasizing focal features, or, conversely, de-emphasizing them in favor of latent features (Fu et al., 2022). Attention and difference-based feature manipulation via the introduction of an attention branch has been implemented in appearance-based gaze estimation by Murthy and Biswas (2021). More recently, Ren et al. (2023) utilized the attention mechanism with residual networks and a gated recurrent unit (GRU) to achieve the error of 3.66°. However, their network requires sizeable inputs (128 × 128 × 8 eye channel and 256 × 256 × 3 face channel), resulting in a large network size (number of parameters) and computational cost (FLOPs).

Appearance-based gaze estimation approaches that rely on a full face image as convolution input can regress the gaze point with smaller errors than previous models. However, this comes at the cost of a bloated number of parameters. The network's computational complexity hinders its ease of deployment, especially for real time gaze estimation since it demands greater levels of system memory and computational resources, even though the returns are increasingly diminishing. This can be an issue even without face input. As evidence, the numbers of parameters in gaze estimation CNNs count up to 105M (Murthy and Biswas, 2021), 196M (Zhang et al., 2017b) or 848M (Cheng et al., 2020b). For this reason, researchers have identified the need for CNNs with decreased computational costs (Huang et al., 2023).

One avenue for designing a computationally efficient network is by optimizing the required input dimensions. Eye image channels in related works adopt varied crop dimensions (and aspect ratios). Separate eye channels have used the crop size of (i) (Krafka et al., 2016) 224 × 224 px, (ii) (Linden et al., 2019) 224 × 112 px, (iii) (Murthy and Biswas, 2021) 60 × 36 px, (iv) (Chen and Shi, 2019) 64 × 96 px, and (v) (Liu et al., 2021a) 72 × 48 px. Joint eye channel sizes include (i) (Park et al., 2019) 256 × 64 px and (ii) (Guo et al., 2021) 256 × 64 px. A lack of research on the impact of the selection of eye crop size on CNN gaze estimation shows potential for improvement by comparing crop sizes to find a size representing agreeable tradeoff between minimizing size and preserving essential information.

A gap that the proposed network seeks to fill pertains to the perceived opportunity for a convolutional network with the gaze estimation mechanism to estimate gaze more efficiently (fewer FLOPs and parameters), but without a cost to accuracy. By reducing the dimensions of the required inputs, a leaner architectural design is sought.

Table 1
CNN architectures for appearance-based gaze estimation without calibration, including the proposed RAGE-net architecture that outperforms them. Other recent works utilize the face input, which comes at the cost of heightened network complexity. Other large datasets are utilized for pre-training certain models (* ETH-XGaze, † ImageNet, ‡ Unspecified whether pretraining is leveraged, but involves a network that is typically pretrained with ImageNet by default). Since pre-training generates an accuracy advantage that is decoupled from the network's architecture, angular errors are reported without pre-training where available.

Authors	Year	MPIIFaceGaze	MPIIGaze	Network type	Inputs
Eye or face input only					
Zhang et al. (2015)	2015	–	6.3°	LeNet	eyes (60 × 36) and rotation vector
Zhang et al. (2017b)	2017	–	5.5°	VGG16	eyes (60 × 36) and rotation vector
Zhang et al. (2017a)	2017	–	4.8°	CNN with spatial weights	face (448 × 448)
Cheng and Lu (2022)	2022	4.55° (4° pre-trained*)	–	ResNet18 with ViT-Base encoder	face (224 × 224x3)
Hu et al. (2023)	2023	3.89° pre-trained†	–	EfficientNet-B3 with hybrid fine-course classification	face (300 × 300x3)
Leblond-Menard and Achiche (2023)	2023	–	4.5° pre-trained	Irisblocks	eyes (64 × 64x3)
Huang et al. (2023)	2023	–	3.5° pre-trained‡	DenseNet with hierarchical CVAE gaze map generation	eyes (150 × 90)
Presented here	**2024**	**4.08°**	**3.96°**	ResNet18-based, with Batch Normalization and attention branch	**eyes (60 × 36)**
Eye and face input both					
Krafka et al. (2016)	2016	5.6° (evaluated by Zhang et al. (2017a))	–	AlexNet	eyes, face (cropped from original frame 224 × 224x3) and face grid
Chen and Shi (2019)	2019	–	4.8° pre-trained†	Dilated VGG16	eyes (64 × 96) and face (96 × 96)
Zhou et al. (2019)	2019	–	4.18°	CNN extractor, LSTM	eyes and face (224 × 224x3)
Cheng et al. (2020b)	2020	4.3°	–	AlexNet	eyes (60 × 36) and face (224 × 224x3)
Cheng et al. (2020a)	2020	–	4.1°	CNN extractor, attention component	eyes (60 × 36) and face (224 × 224x3)
Murthy and Biswas (2021)	2021	–	4.09°	Dilated VGG, attention branch	eyes (60 × 36) and face (120 × 120)
Murthy et al. (2021)	2021	–	4.3°	Generic dilated CNN	eyes (60 × 36) and face (120 × 120)
Ren et al. (2023)	2023	3.66°	–	ResNet18, with AC and GRU module	eyes (128 × 128x8) and face (256 × 256x3)

4. Proposed architecture

We designed RAGE-net (Residual Attention-based Gaze Estimation network), a novel CNN-based architecture with the objective of gaze point prediction that performs comparably to state-of-the-art calibrationless models with the attention mechanism (Murthy and Biswas, 2021; Ren et al., 2023), while preserving low computational resource demands (FLOPs, number of parameters). RAGE-net then serves as the foundation of our evaluation of CNN-based models with similar accuracy in constrained 2D screen environments, including the analysis of the effects of environmental factors on gaze prediction.

RAGE-net is defined by the following core architectural principles (see Fig. 2 for a schema of the architectural model):

- The CNN backbone is based on the ResNet18 architecture, with batch normalization layers incorporated to accelerate convergence, thus allowing for a deeper network where the problem of network degradation has been solved.
- The face channel is excluded, reducing size of the model (28 million parameters) and computational complexity. Eye channels cropped from face images serve as network inputs. A normalization pipeline is followed to process the input images.
- Parallel feature extraction and attention branches, inspired in principle by the successful use of the attention approach for gaze estimation by Murthy and Biswas (2021). Fully-connected layers are optimized via hyperparameter tuning.

RAGE-net is tuned with MPIIFaceGaze to incorporate real user data with realistic illumination and camera angle conditions. To achieve high efficiency in the ratio of size and performance, design principles listed below are implemented. The summary includes variants that were evaluated in parallel during ablation, thus postulating an optimized architectural proposal.

Residual network. For the fundamental framework of the branch structure in both the feature extraction and the attention branch, the convolutional layers of the residual network ResNet18 are adopted. Reuse of activation functions in the ResNet layers can enhance the efficiency of feature learning and reduce redundancy, resulting in improved overall performance (He et al., 2016). Convolutional part of regular ResNet18 has been previously employed for appearance-based gaze estimation by Linden et al. (2019).

Intra-branch weight sharing. As a useful optimization method aimed at decreasing the number of parameters in the ResNet layers by half, RAGE-net takes advantage of the similarity and symmetry between features of the left and right eye. The same instance of the attention and feature branch is utilized to share weights between the two eye channels. Additionally, this approach functions as regularization, decreasing the chance of overfitting to small differences between sides of the eye. Skip connections in the ResNet architecture enable the reuse of features in deeper layers, promoting the sharing of learned representations. Owing to intra-branch weight sharing, RAGE-net is not reliant on parameter reduction strategies such as reducing the input to a single eye channel. Doing so could further reduce computational cost (Huang et al., 2023), but at the potential expense of resistance to variability, or a missing eye channel (e.g., due to uneven illumination) in realistic datasets such as MPIIFaceGaze. Missing views are a common problem in real-world applications (Wang et al., 2024). Eye channel cropping is RAGE-net's primary FLOPs reduction strategy.

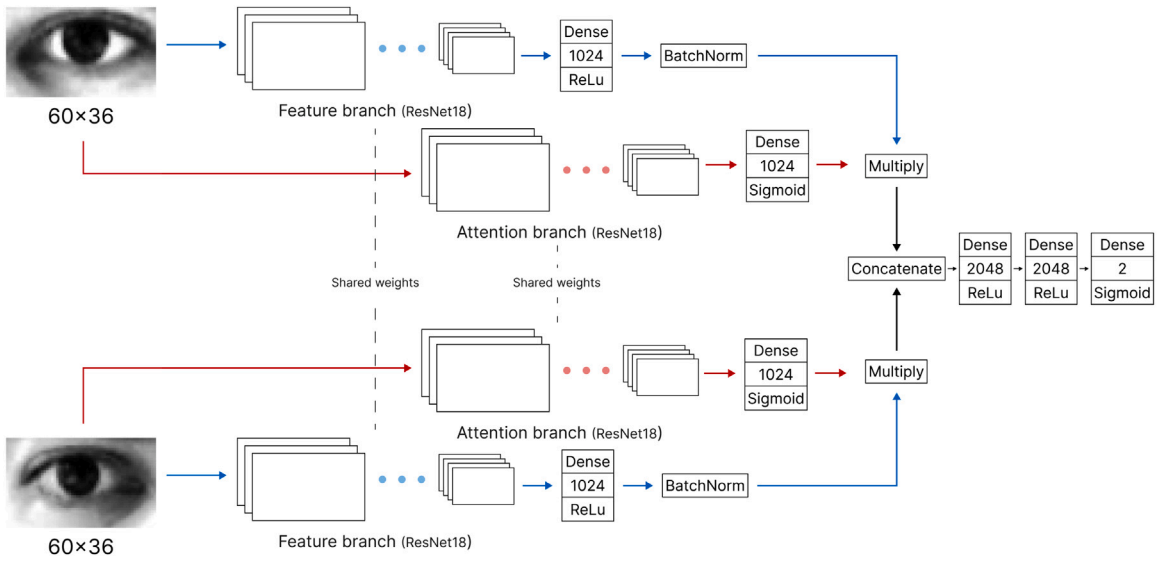


Fig. 2. RAGE-net architecture diagram — Left and right eye channels are passed into feature and attention residual convolution network branches. Attention principle is applied by multiplication between the branches. Dense layers' unit numbers and activation functions have been tuned.

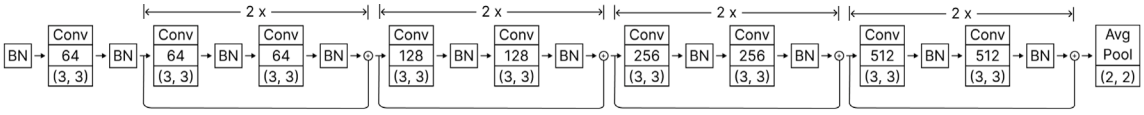


Fig. 3. Network diagram of the convolutional part of RAGE-net's feature and attention branches. Base convolutional section of ResNet18 is enhanced with batch normalization layers preceding convolutions. The structure is identical between the two branches. Intra-branch, weights (convolution kernels) are shared between the left and right eye channels.

Batch normalization. In the feature and attention branches, the base ResNet convolution section of ResNet is enhanced by the addition of batch normalization layers in front of each convolution layer. See Fig. 3 for an overview of the feature and attention branch structure enhanced with batch normalization. Batch normalization is also used after the post-convolution dense layer in the feature branch. Its use facilitates faster learning in a ResNet-based deep neural network. The contribution of incorporating batch normalization after other dense layers was assessed, but yielded no improvement.

Input image normalization pipeline. The goal of normalization is to facilitate uniform treatment of input images acquired in various real-world conditions. The camera's distortion coefficients are used to undistort the input. During further preprocessing of inputs for appearance-based gaze estimation, the normalization approaches are twofold—pose normalization as described by Zhang et al. (2017a) and histogram equalization. The concept of pose is defined as the rotation and translation of the head's coordinate system with respect to the camera coordinate system. For pose estimation, a mapping of 2D facial landmark points to 3D generic facial landmark points is used, calculated with the camera's calibration (projection matrix and distortion coefficients). To detect the 2D facial landmark points, a combination of the OpenCV DNN Caffe model and Dlib is employed for face detection and landmark annotation respectively. Histogram equalization of grayscale eye images boosts their contrast, mitigating the impact of illumination. See Fig. 4 for a flowchart of the image normalization pipeline. The final architecture utilizes pose normalization with histogram equalization.

Eye channel crop size variants. Small dimensions of eye cropping are useful for decreasing the input size to lower computational complexity (a key aim of RAGE-net). Images of the two eyes are cropped from the normalized face image, sized 60 × 36 px. This crop size is selected as yielding the highest accuracy, compared to smaller evaluated crop sizes and crop windows covering both eyes (see 5.1 Experiment 1: Ablation study). Due to the normalization used, the size of the face from which eyes are extracted remains constant. Changing the crop size does not refer to scaling the crop—as would be the result of linear interpolation to achieve the same image of different size—but rather capturing a broader area around the eyes.

No facial channel. Exclusion of the face channel reduces the size of the network. This also results in the CNN losing access to data outside eye appearance. With the gaze direction being a function of head pose and rotation of eyeballs in reference to the head, it is possible that exclusion of the face channel may lead to loss of prediction task performance due to information loss. This is a tradeoff to achieve reasonable accuracy at significantly smaller computational complexity and training time.

A compromise solution including facial information was evaluated as part of the design process - a face grid (Krafka et al., 2016) with the position of the face within a 25 × 25 binary mask. However, it was found to have negligible impact on gaze estimation's accuracy.

Tuned fully connected layers. The convolutional section of the feature extraction branch is connected to a 1024 ReLu fully-connected layer outputting into a batch normalization. The attention branch's convolutional section output flows into a 1024 fully-connected Sigmoid layer (to produce attention in a 0–1 range), without added batch normalization. After the feature and attention branch pass through multiplication and the outputs for both eye channels are concatenated, they are fed to a sequence of three fully-connected layers — double 2048 ReLu and the output Sigmoid returning a zero-to-one-scaled X and Y coordinates of the 2D gaze point on the screen.

5. Experiments and results

Three experiments were conducted as part of our study involving the proposed appearance-based gaze estimation convolutional neural network (see Fig. 5). The first is the ablation study (Experiment 1) conducted as an empirical foundation of the proposed architecture. The second is an experiment (Experiment 2) where results of appearance based gaze estimation are applied to generate a heatmap for the purposes of usability evaluation. Qualitative and quantitative comparison

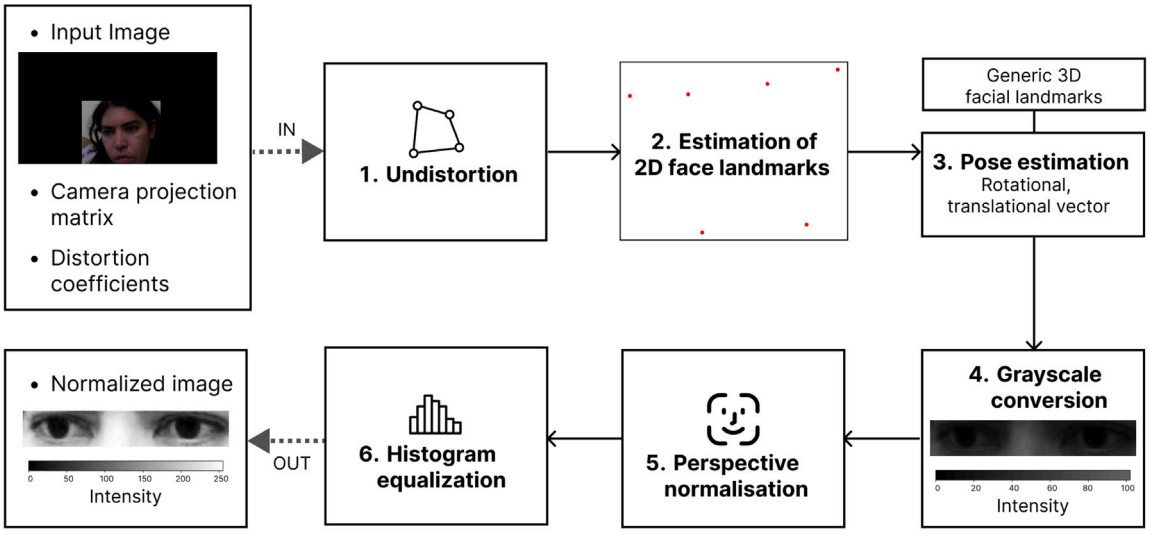


Fig. 4. The RAGE-net input image normalization pipeline starts with removal of camera distortion (1). Six facial landmarks are located (corners of mouth and of both eyes) (2) to allow for pose estimation (3). The image is converted into grayscale (4) before transforming the perspective (rotation and transition) based on the pose (5). Histogram equalization (6) normalizes the image's contrast level.

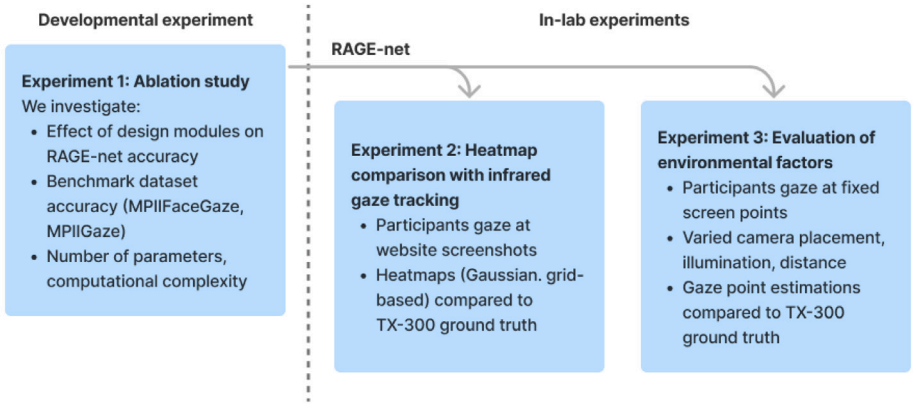


Fig. 5. The methodology of our study comprises three experimental procedures: an ablation study of the proposed RAGE-net network, which yields the network to analyze applicability of appearance-based gaze estimation.

with infrared feature-based gaze estimation is performed. The third (Experiment 3) is an experiment where specific environmental conditions are further examined for their impact on appearance-based gaze estimation — illumination, camera placement, distance of the user from the camera and whether they are wearing glasses.

5.1. Experiment 1: Ablation study

Ablation is conducted to comprehensively evaluate a variety of architectural aspects incrementally modified to design the proposed network. Input face images are sourced from the MPIIFaceGaze dataset (Cheng and Lu, 2022; Zhang et al., 2017b,a), a common benchmark for appearance-based gaze estimation. MPIIFaceGaze supplies data from 15 participants periodically gazing at sequences of points under realistically highly varied lighting and pose conditions. To evaluate the performance of the final network, two datasets are used: MPIIFaceGaze and MPIIGaze (Zhang et al., 2017b).

VGG16, VGG19, and ResNet18 were considered as the core of the network. To determine the best option, base versions of the networks were implemented with separate left and right eye input sized at 60 × 36 pixels (a common eye crop size in appearance-based gaze estimation summarized in Table 1) and lacking any additional attributes. ResNet18 arrived at the best mean squared error (MSE) at .034 (see Table 2). Choosing ResNet18 as the best option, its performance can be further optimized to average MSE = .031 by inclusion of batch normalization in front of each of its convolutional layers.

Decreasing the size of the input can decrease the number of network's parameters. The popularly used eye crop size 60 × 36 px encompasses the surrounding area, including potentially the bridge of the nose and pose of the orbicularis oculi muscle. We investigate whether capturing the eye's surroundings justifies the greater input size, and whether split or joined eye inputs are better. For ablation, eyes were separated into two individual images: (i) 60 × 36 px, (ii) 50 × 36 px, and (iii) 46 × 26 px, or into a single joint image capturing both eyes: (i) 160 × 36 px and (ii) 150 × 26 px. Results provide evidence of 60 × 36 px as worth maintaining as the best crop size, with an average MSE of .036 (see Table 2).

Pose normalization (as adapted from Zhang et al. (2015)) is evaluated for its contribution to gaze estimation due to possible contrasting views on normalization of input data in a CNN. Pose normalization can eliminate the noise of varied camera positioning and scaling found in the wild to arrive at unified representation of eye inputs. Interpreting this alternatively, the pre-normalization state of the eye input may itself carry information about direction of the gaze. At MSE of .037, the version of the network with pose normalization performs better than without it (.042).

The presence of a face grid input (a 25 × 25 binary mask describing the position of the face in the source camera image) is examined.

Table 2
Results of the ablation study conducted to support design decisions made in the proposed network RAGE-net. Selected options with lowest error are highlighted.

Trial number	CNN network	Eye crop	Pose normalization	Grid input	Attention branch	MSE (avg. over 3 runs)
Ablation of CNN networks						
1	VGG16	60x36	Yes	–	–	.036
2	VGG19	60x36	Yes	–	–	.053
3	ResNet18	60x36	Yes	–	–	.034
4	ResNet18 with batch normalization	60x36	Yes	–	–	**.031**
Ablation of crop size						
5	VGG16	60x36 split	Yes	–	–	**.036**
6	VGG16	50x36 split	Yes	–	–	.041
7	VGG16	46x26 split	Yes	–	–	.044
8	VGG16	160x36 joined	Yes	–	–	.045
9	VGG16	150x26 joined	Yes	–	–	.050
Ablation of pose normalization						
10	VGG16	60x36	Yes	–	–	**.037**
11	VGG16	60x36	–	–	–	.042
Ablation of grid input						
12	ResNet18 with batch normalization	60x36	Yes	Yes	–	.033
13	ResNet18 with batch normalization	60x36	Yes	–	–	**.031**
Ablation of attention branch						
12	ResNet18 with batch normalization	60x36	Yes	–	–	.031
13	ResNet18 with batch normalization	60x36	Yes	–	Yes	**.029**

Pose normalization and removal of the face image input can raise concerns that the network discards key information about the head for more effective prediction of gaze direction. Addition of the face grid was measured to have no direct benefit, even slightly decreasing gaze estimation accuracy (see Table 2).

Finally, comparison of the networks with and without the attention branch validates the inclusion of the mechanism to improve the network's error (see Table 2).

The most effective variants (ResNet18 with batch normalization, 60 × 36 split eye crop size, with pose normalization and the attention branch and without face grid input) were compiled into an initial model of the realized network, utilizing an attention branch and weight sharing between eye branches. The batch size was set to 32. Early stopping regularization was used from epoch 10, with minimum delta of $1e^{-4}$ and patience parameter of 2. Hyperparameter tuning was conducted with this architecture with the following hyperparameters:

- Cyclical learning rate (LR) changed linearly within a range from the minimum value of $5e^{-6}$ to the maximum value, determined individually for each set of hyperparameters by using LR range test - 4 variants - [$1e^{-4}$], [$5e^{-2}$], [$7.5e^{-2}$], [$1e^{-1}$])
- Number of units in the dense layers within the feature and attention branch, immediately after the ResNet18-like convolution layers (5 variants - [256], [512], [1024], [2048], [9096])
- Number of units in the dense layers following the concatenation of left and right eye channels (6 variants - [256, 512], [2048, 1024], [2048, 2048], [3072, 3072], [4096, 2048], [4096, 4096])
- Use of additional batch normalization after each dense layer past concatenation (2 variants - [enabled] or [disabled])

The optimal hyperparameters were found to be (as implemented in Fig. 2):

- Cyclical learning rate: [$5e^{-6}$, $1e^{-4}$]
- Number of units in the dense layers within the feature and attention branch: **[1024]**
- Number of units in the dense layers following the concatenation of left and right eye channels: **[2048, 2048]**
- Additional batch normalization after dense layers past concatenation: **[disabled]**

The cross-person angular error of our final hyperparameter-tuned network (RAGE-net) is evaluated. In MPIIFaceGaze, the result of 15-fold cross-validation yields the error of 4.08°. MPIIGaze provides an evaluation subset, which however contains data from the same users as the learning subset. To avoid subject-specific bias, leave-one-person-out cross validation is performed instead, yielding the angular error of 3.96°.

Computational costs of RAGE-net show that the network also succeeds at being light-weight and efficient. The total number of parameters in the network is 28.66M, and the number of GFLOPs is 9.79. This is a 73% reduction compared to the 105M parameters in the closest related work by Murthy and Biswas (2021). These values are comparable to other high-efficiency networks such as the VGE-Net (Huang et al., 2023), which has 9.8M parameters and 71.3 GFLOPs in a configuration that optimizes the balance between complexity and accuracy. While RAGE-net's number of parameters is higher, its computational efficiency makes it valuable for real-time gaze estimation, reducing the load on the processing units to not affect the user experience of other software in which the user's gaze is being observed.

5.2. Experiment 2: Heatmap comparison with infrared gaze tracking

Angular error (Eq. (1)) reported as a metric of performance of gaze estimation methods may be seen as non-indicative of the actual practical accuracy of a method that the degrees represent. Besides making comparisons with other gaze estimation methods, its values are difficult to interpret (What angular error is negligible? What error is too high?).

For the purpose of human interpretation of gaze tracking, the objective of this experiment is the comparison between outputs of our appearance-based RAGE-net model and a commercially available infrared feature-based gaze tracking system. The experiment was conducted in compliance with the GDPR, with participants asked for their explicit consent before any data collection.

A controlled experiment was conducted in a lab. The computer operated by the participant was equipped with TX300, a 23″ 1920 × 1080 monitor with a built-in 300 Hz infrared eye tracker, and Creative Senz3D, an off-the-shelf external webcam with a 24 Hz frame rate and the output resolution of 1920 × 1080. For direct comparisons of methods utilizing data from the two sensors, both the eye tracker and the webcam were deployed to monitor the participant in parallel. Tobii Pro Lab is the software used to obtain the results of infrared gaze tracking. Webcam data is received by RAGE-net, while the corneal reflection technique employed by Tobii's infrared gaze tracking serves

Table 3
Cross-person comparison of Gaussian and grid heatmaps of individual stimuli. Measures are the Jensen–Shannon divergence (JSD), bounded on a [0,1] interval and structural index similarity (SSIM), on a [−1,1] interval. For identical heatmaps, JSD would be 0 and SSIM would be 1.

Stimulus ID	Gaussian - SSIM	Grid - JSD (SSIM)						
	1920 × 1080	**4 × 3**	**5 × 3**	**5 × 4**	**8 × 4**	**10 × 8**	**13 × 10**	**16 × 9**
1	0.45	0.10, (0.68)	0.11, (0.69)	0.11, (0.71)	0.14, (0.72)	0.29, (0.75)	0.30, (0.84)	0.33, (0.84)
2	0.59	0.19, (0.63)	0.19, (0.66)	0.24, (0.61)	0.25, (0.66)	0.33, (0.77)	0.34, (0.84)	0.35, (0.85)
3	0.53	0.09, (0.70)	0.10, (0.68)	0.15, (0.67)	0.17, (0.70)	0.32, (0.77)	0.29, (0.86)	0.30, (0.87)
4	0.69	0.16, (0.73)	0.17, (0.70)	0.20, (0.72)	0.21, (0.77)	0.28, (0.81)	0.28, (0.86)	0.29, (0.86)
5	0.62	0.15, (0.65)	0.15, (0.68)	0.21, (0.64)	0.22, (0.71)	0.30, (0.79)	0.33, (0.86)	0.32, (0.86)
6	0.62	0.04, (0.82)	0.08, (0.76)	0.19, (0.63)	0.22, (0.71)	0.35, (0.79)	0.37, (0.84)	0.37, (0.85)

as an industry-standard reference of a feature-based gaze estimation approach, supplying the ground truth.

Twelve (12) participants, comprising 10 men and 2 women aged between 21 and 38 years old, were recruited for the study in Slovakia as a convenience sample.

Participants are presented with website screenshot stimuli for 10 s each, with every stimulus followed up by questions in the style of a first impression test (Kuric et al., 2023). Screenshots of six real websites are presented. Questions are asked by the moderator to avoid use of a mouse and keyboard and to achieve greater consistency of head poses. The questions pertain to the brand name and value proposal (type of service/product and the reason for selecting it).

Data that corresponds to each task's 10-second time interval was extracted from the eye tracker and the webcam. Camera footage was converted into a sequence of jpeg images. As part of the input image normalization pipeline, the OpenCV[4] Caffe model is used to detect the face bounding box and Dlib to locate the facial landmarks. The camera's projection matrix and distortion coefficients, which are utilized for mapping 2D coordinates from an image to 3D coordinates, were obtained using the OpenCV library. The function OpenCV.findChessboardCorners was employed to identify the corners of 5 × 7 chessboard across 13 images, while OpenCV.calibrateCamera was used to derive the extrinsic calibration parameters. The RAGE-net model is applied for gaze estimation. An I-VT fixation filter was implemented to obtain fixations from the estimated gaze points.

Heatmaps are normalized 2-dimensional probability distributions. Therefore, for their robust comparison that takes into account local differences, the Jensen–Shannon divergence is used as a measure of comparison. Jensen–Shannon divergence (JSD) (Chen et al., 2023) is a commonly used symmetrical and bounded mathematical measure of difference between probability distributions. Additionally, since heatmaps are commonly visualized as images, the structural index similarity (SSIM) (Setiadi, 2021) is used as a measure that models similarity according to the human visual perception of images.

For explorative analysis of the qualitative nature of fixations derived from appearance-based gaze estimation and their comparison with infrared gaze tracking in full detail, Gaussian heatmaps were calculated. Visual similarity of heatmaps is stimulus dependent (see Table 3), ranging from moderately low (0.45) to moderately high (0.69). However, qualitative visual comparison reveals the similarity as inflated by wider spatial distribution of areas without fixations (see Fig. 6). In comparison between webcam-based (RAGE-net) and infrared-based gaze estimation heatmaps, the former is not accurate enough to pinpoint specific user interface elements that receive most visual attention. The general area of visual attention matches with infrared gaze tracking, but none of the focused elements can be identified). The estimated fixations are diffused over the surrounding area, including white spaces that received no actual attention during the experiment.

Given the dissimilarity found between Gaussian heatmaps, as an alternative representation, grid-based heatmaps with varied degrees of granularity are investigated to gauge the similarity in fixations between webcam-based and infrared gaze tracking. The less accurate appearance-based method could be perceived as providing satisfactory results if the fixation is found mapped in the accurate cell. Evaluated heatmap meshes divide the screen into areas of interest by following ratios: 4 × 3, 5 × 3, 5 × 4, 8 × 4, 10 × 8, 13 × 10 and 16 × 9.

Across all stimuli, grid-based heatmaps have higher SSIM than Gaussian heatmaps (see Table 3), which demonstrates that when interpreted at a lower level of granularity, appearance-based gaze estimation can become a more reliable visual approximation of the ground truth. JSD is the lowest at 4 × 3 and highest at 16 × 9, corroborating that high angular errors have lesser impacts on coarser grids. The positive association between JSD and SSIM in grid heatmaps can be explained by the cells in coarser heatmaps covering larger areas of the image, granting differences in individual cells a greater impact on appearance of the heatmap. There is variance in how closely the grid from the webcam-based approach resembles the TX-300 between stimuli (see Fig. 7) and between individuals. Cross-stimulus average JSD calculated per-person ranges between .10–.45 for the course-grained 4 × 3 heatmap grid and .38–.65 for the finest-grained 16 × 9 heatmap grid.

5.3. Experiment 3: Evaluation of environmental factors

Various factors of the environment – distance, illumination, angle of the camera or presence of glasses – affect the eyes' appearance as captured through the lens of the webcam. RAGE-net's (our architecture's) angular error as a measure of performance is reasonably comparable to state-of-the-art CNN results, as shown in Table 1. However, it does not indicate how said performance can be affected by the environment in further detail.

Previous works have explored the effect of some environmental factors (Zhang et al., 2019; Murthy et al., 2021). Wearing spectacles and being outdoors were found to increase the error of gaze estimation, while distance between the user and the camera was found to have no negative impact. The aim of our environmental factor evaluation is as follows:

- To investigate the effects of alterations of the environment's independent variables specifically for RAGE-net, a different network, potentially replicating some results demonstrated by Zhang et al. (2019), thus contributing to the knowledge of whether such findings could be generalizable as a theory for other CNN gaze estimation solutions.
- To expand the scope of investigated environmental factors. For illustration, illuminance outdoors is around 100–1000 times higher than indoors (Bhandary et al., 2021). Can the smaller indoor variability in illumination (the desktop computer's most common environment) still affect gaze estimation accuracy? How is gaze estimation affected by different positioning of the webcam?
- To perform further exploratory and qualitative analysis on top of testing hypotheses about the effect on a numerical measure of accuracy. We posit that understanding the manner in which the environment changes the gaze predictions, and describing these problems can lead to the development of more accurate gaze estimation methods.

[4] OpenCV: https://opencv.org/.

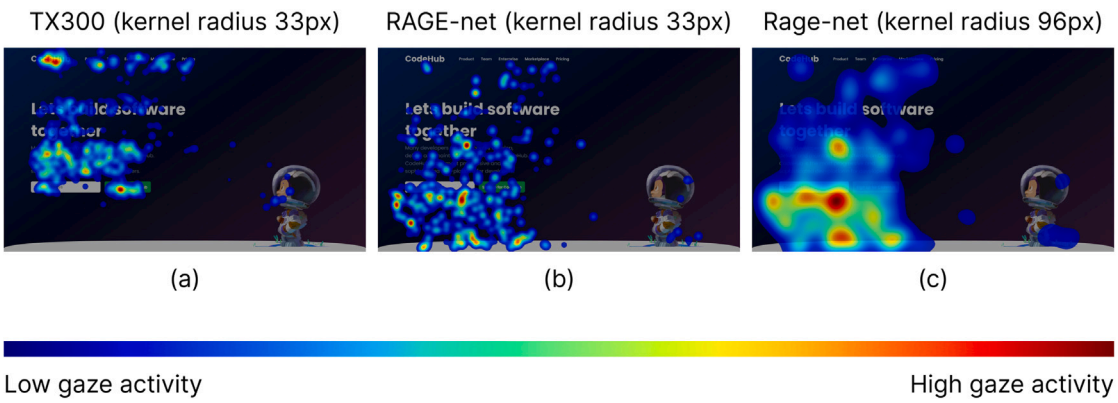


Fig. 6. Comparison of gaze heatmaps for the same pageview estimated by feature-based infrared approach of (a) the TX300 eye tracker and (b) appearance-based RAGE-net proposed here. Visual attention spent on specific user interface elements visible in TX300 is distinctly lacking In RAGE-net. Due to the inaccuracy of the appearance-based approach, (c) - an alternative version of the heatmap is presented, where the heatmap is plotted with a large Gaussian kernel radius of 96px, to illustrate a higher angular error of the gaze estimation.

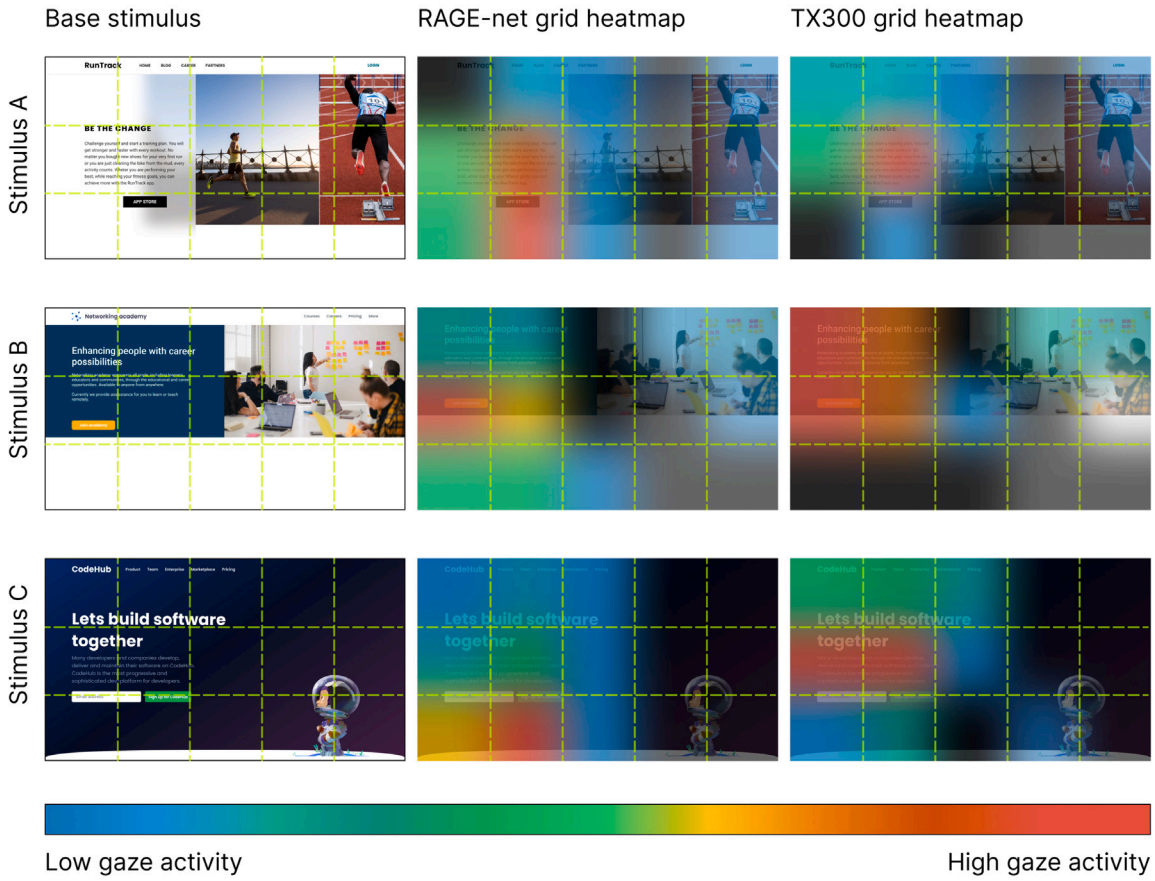


Fig. 7. Comparison of grid 5 × 3 heatmaps of appearance-based gaze estimation (proposed RAGE-net network) with highly accurate feature-based gaze tracking (TX300). The grid layer is blurred around grid borders to communicate that gaze predicted by appearance-based approaches does not always perfectly fit within cells due to lower accuracy.

Therefore, we ask the following research question:

RQ: How do the factors of the environment independent from gaze direction affect the accuracy of appearance-based gaze estimation when compared to feature-based (infrared) eye tracking?

After identifying relevant–commonly appearing–environmental factors, hypotheses are stated as thus:

- **H1 Glasses: The presence of participants wearing corrective dioptric glasses increases the error of the proposed network similarly to feature-based gaze estimation.** *Justification: Light reflection, the frame blocking view of the eye and interference with facial landmark detection are some of the causes for why glasses can be expected to have a negative impact on both gaze estimation approaches.*
- **H2 Increased distance: Changing the face-screen distance from the optimal viewing distance increases the error of the proposed network similarly to feature-based gaze estimation.** *Justification: Decrease in accuracy can be expected for both approaches. TX300 documentation declares increased error past the*

65 cm distance benchmark.[5] *A reasonable forecast can be made that due to the scaling step while mapping a 2D face image to a generic 3D face model, greater differences in distance will cause a decrease in accuracy.*

- **H3 Webcam placed below the screen: Placement of the webcam below the screen increases the error of the proposed network.** *Justification: Webcams are commonly (though not necessarily) placed on top of the screen. It is plausible to expect that the training dataset will be biased to perform worse if the webcam is placed below the screen. Statistical comparison with feature-based gaze estimation is not performed, since it is more typical for an infrared eye tracker to be placed in a standardized position (e.g., the infrared camera in the TX300 is monitor-mounted, situated below the screen), while webcam placement may vary more in the wild, depending on user preferences.*
- **H4 Webcam placed on the left: Placement of the webcam on the left side of the screen increases the error of the proposed network.** *Justification: Analogically to justification of H3, but with the placement of the webcam on the side of the screen. Findings can be presumed to apply symmetrically with mirrored placement of the webcam on the right side of the screen.*
- **H5 Darkness: Low illumination increases the error of the proposed network more than in feature-based gaze estimation.** *Justification: Appearance-based gaze estimation is dependent on presence of visible-spectrum light in the camera's scene. Lack of illumination can be expected to impact the prediction model's ability to estimate the gaze point.*
- **H6 Harsh light: High illumination increases the error of the proposed network more than in feature-based gaze estimation.** *Justification: Analogically to justification of H5, overabundant illumination can be expected to accentuate the error of gaze estimation.*

Existing datasets used in gaze estimation research, such as MPIIFaceGaze include a variety of environmental factors in the recorded scene. However, given their origin in an uncontrolled experiment, the values of individual factors (e.g., the degree of illumination) are not fully documented or equally represented in the sample. Therefore, a new experiment was designed and conducted to obtain data in controlled conditions.

The experiment consists of six analogous blocks, each using a different set of environmental factors. During each block, 27 points are displayed to the participants (3 sets of 9 positions fixed in a grid). The order of positions is randomly generated. Participants are instructed to gaze on each point after it appears, then to press the left mouse button to proceed to the next point. Between the blocks, an empty screen is displayed while the moderator adjusts the environmental factors to start the next block.

The same technical setup (monitor-mounted eye tracker, webcam) is used as in experiment 2. Six environmental conditions are designed, including a single reference condition and five alternative conditions, each differentiated from the reference by a single factor. The ideal distance from a 23" full HD monitor (70 cm) is compared with an alternate viewing distance that is greater yet realistic (80 cm). After the required initial calibration, the TX300 eye tracker is not recalibrated between conditions. This is to judge its performance in conditions similar to the proposed network, which functions without calibration. Based on its documentation, this is expected to decrease its accuracy when illumination changes due to dilation/constriction of the participant's pupil.

Differences in the illumination factor are produced by inclusion of two light sources — ambient light produced by a fluorescent light ceiling fixture with multiple power options, and front ring light (Rollei Lumis Tabletop Bi-color). Natural light is completely blocked off by window blinds, assuring consistent artificial lighting conditions for every participant, regardless of the time of day. The laboratory setting, including the ring light, light fixture illumination and blinds, is shown in Fig. 8. The conjunction of a light fixture set to medium power and front lighting is expected to provide ideal lighting conditions for gaze estimation. Conditions where (i) the monitor is the only source of light and (ii) a light fixture is set to high power, are used to emulate suboptimal lighting conditions (a dark room and harsh office lighting respectively). Illuminance is measured with a lux meter, at the distance of one 1 m from and facing the light source (ceiling-mounted light fixture and monitor respectively). For further specifics of the conditions, see Table 4.

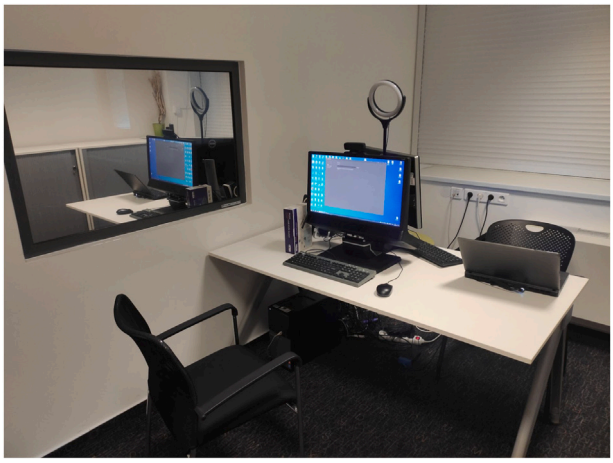


Fig. 8. Computer setup during the experiment for evaluating environmental factors' impact of gaze estimation (pictured: front ring light, webcam on top of the screen, shut blinds to control illumination by removal of natural light from the scene).

Thirty (30) people were recruited in Slovakia through convenience sampling. Participants are aged 20–35, with gender distribution 27 men and 3 women. Seven (7) participants wear dioptric glasses and two (2) participants wear contact lenses. For the purposes of our evaluation, people with contact lenses are treated as part of the group "without glasses" (though contact lenses have also been linked to lowered eye tracking accuracy Holmqvist et al., 2023). See Fig. 9 for illustration of the appearance of the eyes with and without glasses in varied environmental conditions, where glasses and their glare can be seen obstructing the eye.

Between individuals, gaze estimation via both approaches performs with considerably variable accuracy. As shown in Fig. 10, TX300 captures the gaze of the best-estimated participant with high precision, close to the ground truth gaze points. With the worst-estimated participant, some noise appears, though clusters around real gaze points are still recognizable. With RAGE-net, the gaze of the best-estimated participant is clustered around the real gaze points, despite the expected lower accuracy. For the worst-estimated participant, gaze clusters cannot be clearly visually identified, even missing from certain gaze points (middle).

Hypotheses H2-H6 are primarily investigated using the sample of 23 participants without glasses, to assess the impact of environmental factors independently from the potential effect of glasses. The complete 30-participant dataset is also evaluated separately, to check whether the presence of glasses causes any notable changes. In the unaltered condition 1 - Reference, the gaze of all participants clusters in elliptical areas surrounding the real gaze point, as shown in Fig. 11. This yields the condition as a suitable benchmark for environmental factor conditions to be compared to.

[5] Tobii Pro TX300 documentation: https://www.spectratech.gr/Web/Tobii/pdf/TX300.pdf.

Table 4
Reference condition (1) uses medium-powered fluorescent light fixation and front lighting, at 70-cm distance from the monitor and camera placed on top of the monitor. Alternative conditions differ by one factor - (2) 80-cm distance from the monitor, (3) camera placed on the bottom of the monitor and (4) left of the monitor, (5) monitor as the only light source and (6) harsh light provided by high-powered fluorescent light fixation.

Condition title	Lighting	Illuminance facing the ceiling-mounted fluorescent light fixture (lx)	Illuminance facing the monitor (lx)	Distance from the monitor (cm)	Camera placement
1 - Reference	Ceiling-mounted fluorescent light fixture (medium power) + front lighting	125[a]	79	70	Top of the monitor, centered horizontally
2 - Increased distance	Ceiling-mounted fluorescent light fixture (medium power) + front lighting	125[a]	79	80	Top of the monitor, centered horizontally
3 - Webcam placed below the screen	Ceiling-mounted fluorescent light fixture (medium power) + front lighting	125[a]	79	70	Bottom of the monitor, centered horizontally
4 - Webcam placed on the left	Ceiling-mounted fluorescent light fixture (medium power) + front lighting	125[a]	79	70	Left of monitor, centered vertically
5 - Darkness	Monitor light only	~0	30	70	Top of the monitor, centered horizontally
6 - Harsh light	Ceiling-mounted fluorescent light fixture (high power)	1050	98	70	Top of the monitor, centered horizontally

[a] Due to variance caused by high sensitivity of the light fixture's continuous analog dimmer, mean illuminance is presented.

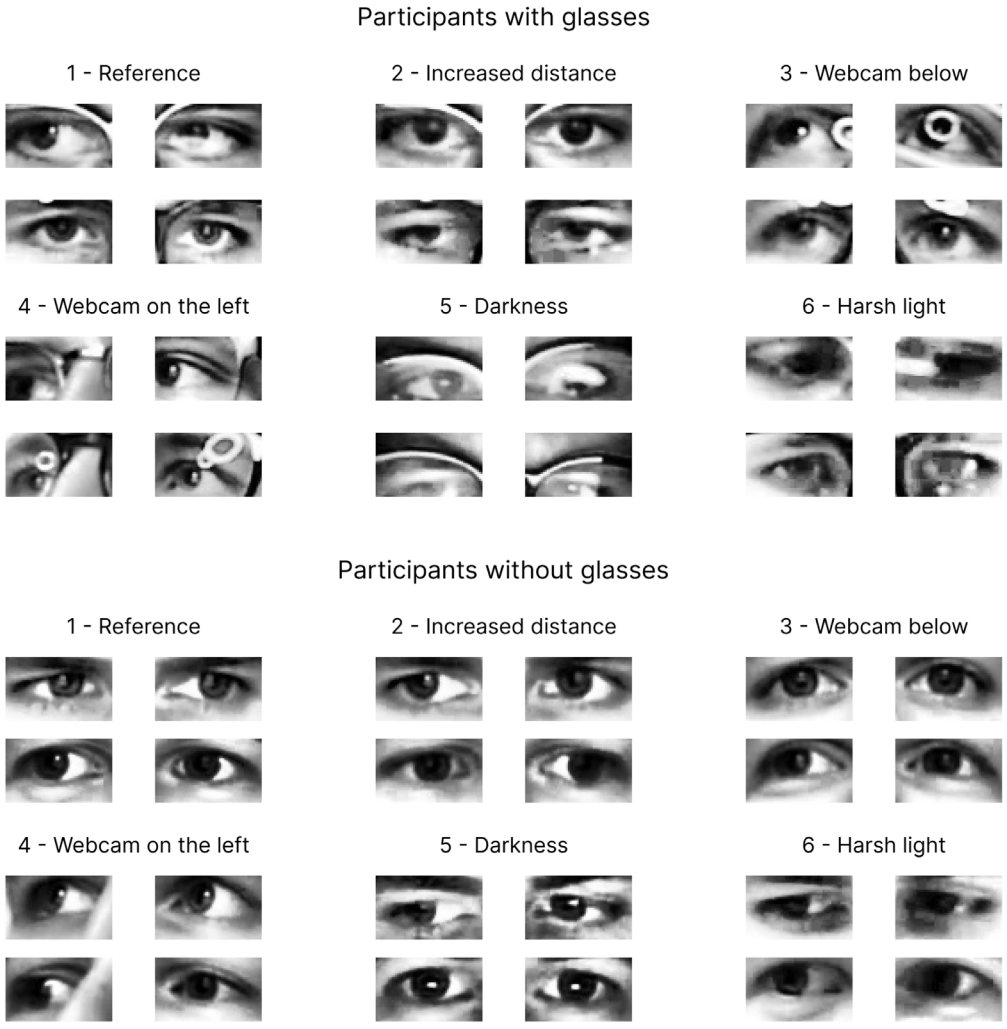


Fig. 9. Cropped eye area input images from experiment 3, gathered in each of the environmental conditions described in Table 4, for participants with and without glasses, pictured post pose normalization and histogram equalization.

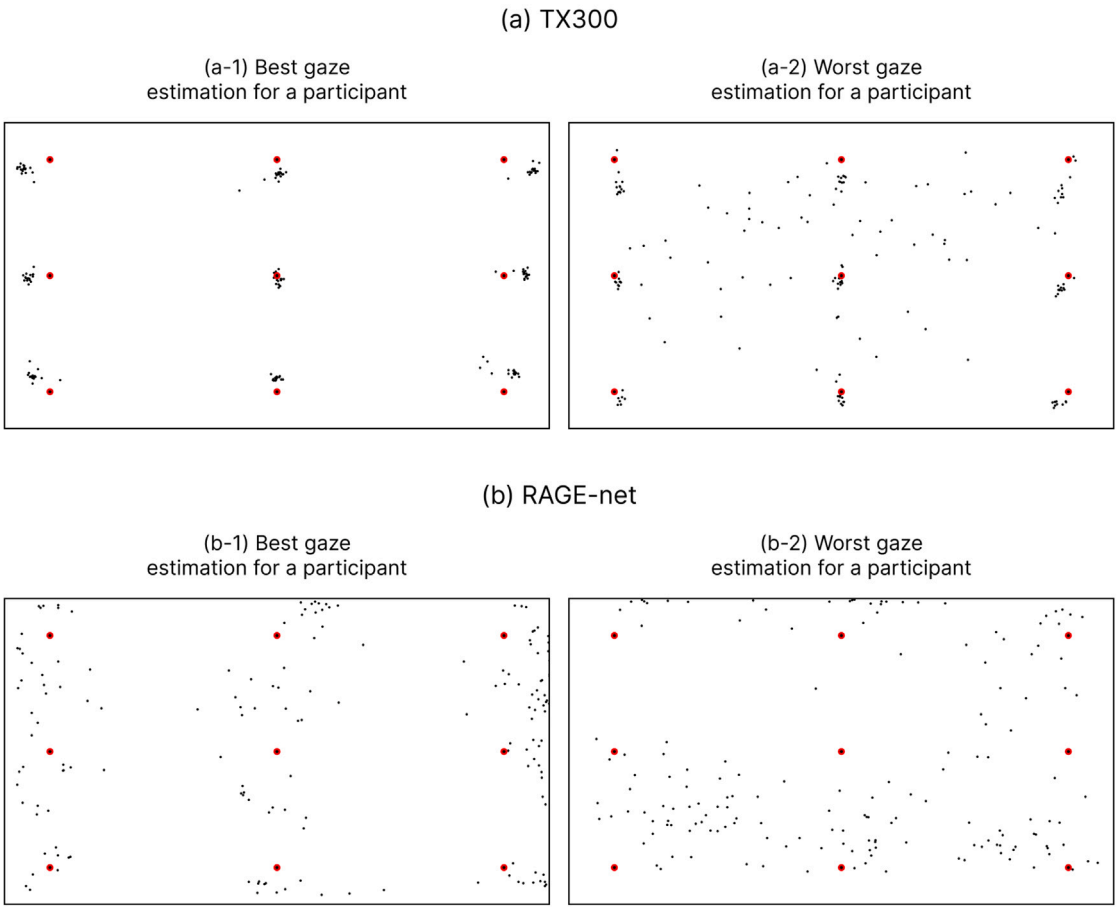


Fig. 10. Between participants whose gaze is estimated with best and worst accuracy across all environmental conditions, the accuracy can vary considerably for both compared approaches (a) and (b).

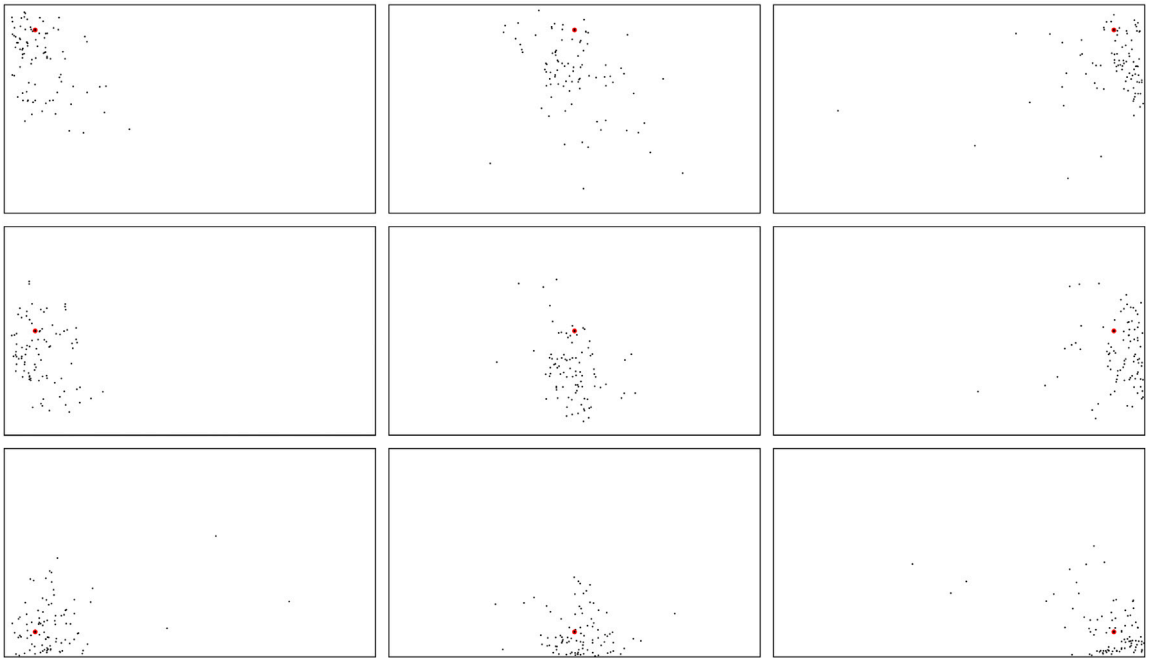


Fig. 11. RAGE-net estimated gaze points from experiment 3 in the 1 - Reference (unaltered) environmental condition. Estimated gaze points are separated between the nine ground-truth gaze positions (red).

As the metric for accepting or rejecting the hypotheses, the on-screen distance error is calculated as Euclidean distance between the expected point of gaze (the actual coordinates of the observed point on the screen) and the estimated gaze point, measured in millimeters. The Shapiro–Wilk test with statistical significance level of .05 is used to test normality of data. The Mann–Whitney U-test with statistical significance level of .05 is used to evaluate the hypotheses.

The N in the nonparametric statistical tests is determined as the product of the number of participants (all: 30, without glasses: 23), the number of observable gaze points per environmental condition (27), and the number of environmental conditions included in the test (hypothesis 1: 6, hypotheses 2–6: 2). Note that the Tobii equipment used for feature-based gaze estimation is a black box, which sometimes neglects to report a gaze point (effectively failing to detect gaze). Therefore, the N in case of feature-based gaze estimation is determined by the actual number of gaze estimations provided by Tobii equipment.

Results indicate that wearing glasses increases the on-screen distance error of feature-based gaze estimation ($z = -7.47, p < .001, r = .11, n = 4488$). Glasses also have a similar impact on the proposed RAGE-net network, ($z = -10.19, p < .001, r = .15, n = 4860$). Hypothesis H1 can therefore be accepted — the effect of wearing glasses is similar between RAGE-net and infrared gaze tracking.

In the proposed network, increased distance from the monitor does not manifest as a statistically significant difference in on-screen distance error ($z = -1.38, p = .084, r = .039, n = 1242$). By contrast, in feature-based gaze estimation, there is evidence of distance having a small effect on the error ($z = -5.05, p < .001, r = .16, n = 1055$). This result is identical for the complete sample (including participants wearing glasses) - no significance of distance for proposed network ($z = -1.31, p = .095, r = .032, n = 1620$), and with small effect in feature based gaze estimation ($z = -5.42, p < .001, r = .15, n = 1314$). Hypothesis H2 is rejected — the impact of distance is dissimilar between the two gaze estimation approaches.

Placement of the camera below the screen is reflected in a significant growth of the on-screen distance error ($z = -19.59, p < .001, r = .56, n = 1242$), regardless of whether participants are allowed to wear glasses ($z = -21.18, p < .001, r = .52, n = 1620$). A similar effect is observed when the camera is placed on the left of the screen ($z = -25.37, p < .001, r = .72, n = 1242$), also replicated if the participants can wear glasses ($z = -28.49, p < .001, r = .71, n = 1620$). Hypotheses H3 and H4 are both accepted — atypical webcam positioning considerably interferes with the proposed network's attempts at gaze estimation. This significant increase of the on-screen distance error is reflected in box plots in Fig. 12. Positions of gaze points in these conditions shift globally, as shown in Fig. 13.

Inducing darkness in the environment has a small effect on the measure of on-screen distance error when using the proposed network ($z = -3.91, p < .001, r = .11, n = 1242$). Darkness also has a lesser effect on feature-based gaze estimation ($z = -2.08, p = .019, r = .059, n = 1226$). When including participants wearing glasses, the result is similar — proposed network ($z = -6.60, p < .001, r = .16, n = 1620$), feature-based gaze estimation ($z = -2.90, p = .002, r = .073, n = 1592$). Hypothesis H5 is accepted — the lack of a light source other than the glow of the monitor impacts the outcome of the webcam-driven approach more than it does infrared gaze tracking, despite attempts at mitigation via the histogram equalization of inputs during data preprocessing.

The effect of harsh light evidenced on the proposed network ($z = -7.90, p < .001, r = .22, n = 1242$), is greater than the effect on feature-based gaze estimation ($z = -3.80, p < .001, r = .11, n = 1232$). The equivalent is true when including participants who wear glasses — proposed network ($z = -7.55, p < .001, r = .19, n = 1620$), feature-based gaze estimation ($z = -3.62, p < .001, r = .09, n = 1600$). Hypothesis H6 is accepted — harsh office light affects gaze estimation via the proposed network more than in infrared gaze tracking.

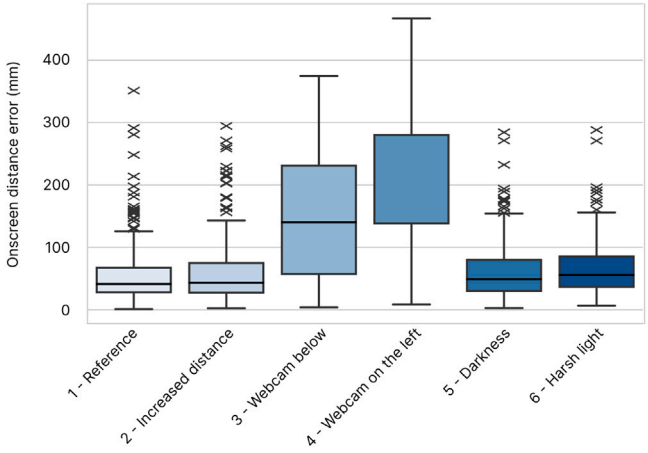


Fig. 12. Box-plots of on-screen distance error for observed environmental conditions (see Table 4) for participants without glasses. The box outlines the upper and lower quartile of the distribution, the box's inner line is the median and the "wicks" represent minimum and maximum values (aside from outliers — the cross marks).

As seen in the comparison of on-screen distance errors between the conditions with reference illumination, darkness and harsh light (see Fig. 12), while significant, their differences are smaller compared to the shift effect that camera placement has. Gaze data in darkness and harsh light manifests as clusters of lower precision around the ground truth gaze point (see Fig. 13).

6. Discussion

With the RAGE-net model presented in this work, we demonstrate that in desktop computer environments, under realistic illumination and angle conditions, it is possible for a residual network with an attention branch that solely utilizes images of cropped-out areas surrounding the eyes as inputs to perform competitively to more complex networks while remaining more computationally efficient (see Table 1 for comparison with previous models). By reducing the size of the input data, the complexity of the network decreases, making the CNN more efficient for practical real-time applications. The significance of this result shines particularly in the light of the fact that a primary motivation for webcam-based approaches is the democratization of gaze tracking. Making gaze tracking broadly accessible to anyone encompasses not just bypassing the need for dedicated infrared cameras, but also requirements for exceptional computational resources. It could be posited that the practical accuracy of the network could be further improved by leveraging the benefits of pre-training with general-task or task-specific datasets such as ImageNet and ETH-XGaze respectively. Though further improvement of RAGE-net's angular error by fractions of a degree could have potential practical value, to contribute to the pursuit of democratization of screen-based gaze tracking in the wild, gaze estimation improvements should be considered in the broader context of our evaluation of the utility of appearance-based gaze estimation, which may bear implications for constructing gaze estimation networks and datasets.

With our model boasting comparable prediction performance and favorably lowered computational complexity to discriminate it against previous related models, it is suitable for the second aim of this research, which is the investigation of the extent to which its results can be indicative of real gaze in praxis. For instance, the angular error of 4.08° in MPIIFaceGaze is comparable to similar approaches, but since the conical field perceived by the fovea is only 0.5° wide (Leigh and Zee, 2015), this raises a valid question, of whether the information from similar appearance-based gaze estimation solutions is a sufficient-enough approximation to represent salient and usable information about real gaze.

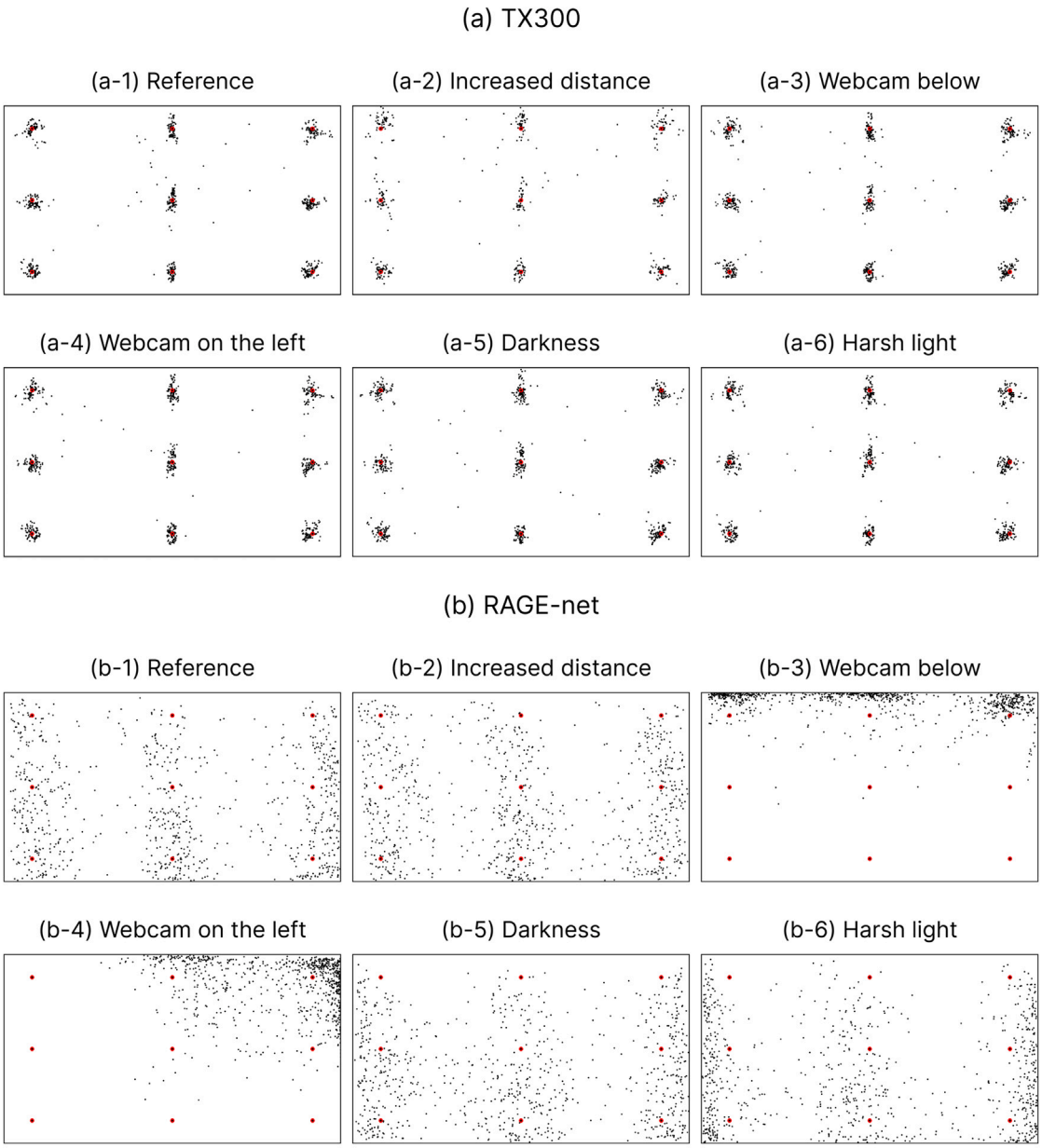


Fig. 13. Comparison of gaze points predicted in experiment 3 via infrared gaze tracker TX300, and webcam-driven appearance-based gaze estimation in environmental conditions listed in Table 4.

The apparent low precision of Gaussian heatmaps obtained from the appearance-based approach (see Fig. 6), bears the implication that at this accuracy, there is a risk of being non-indicative of actual gaze, particularly on screens where user interface elements can be close to each other. It is notable that RAGE-net inputs incorporate the webcam's calibration (distortion coefficients, projection matrix) to transform the gaze to the screen's 2D coordinates. For democratization of eyetracking, in in-the-wild conditions where camera parameters are unavailable (and obtaining them presents a usability barrier), this could lead to further cost to accuracy.

The visualizations of appearance-estimated gaze on the screen should clearly depict its lower level of precision to avoid drawing false conclusions based on misunderstanding of the information being provided. Grid heatmaps could be seen as a form of transparent and unambiguous visual representation of the outputs of appearance-based gaze estimation. The difference between grid heatmaps obtained using RAGE-net and an infrared eye tracker gradually rises with finer level of granularity. Identifying a threshold of maximum granularity at which the heatmaps maintain reliability is subject to interpretation based on the goals of the analysis, and also contingent on the characteristics of the stimulus, as seen in Table 3. In our experiment, the differences become more pronounced above dimensions of 5×3. This corresponds with the 4.08° angular error of the model. On a 23" screen — dimensions of 50.9 cm × 28.6 cm, at the optimal viewing distance of 70 cm, the error corresponds to 4.99 cm. Therefore, by average the fixation should be within a 9.98 cm × 9.98 cm area (4.99 in both directions), approximately matching a 5×3 division of a 23" screen. Therefore, we suggest that to maintain low divergence from actual gaze, cells in grid heatmaps used to represent appearance-based gaze estimation should not have dimensions larger than the estimation method's on-screen distance error.

Examination of grid heatmaps reveals an associated risk caused by gaze shift across neighboring cells. If fixations close to the borders of a cell are subject to a directional shift due to the gaze estimation inaccuracy, estimated heat patterns can deviate considerably from the ground truth. See Fig. 7 for this phenomenon in action when interpreting data from experiment 2. For all pictured stimuli, there is an observable shift in gaze point estimations toward the bottom of the image. The layout of stimulus A aligns most with the grid utilized, resulting in heatmaps that are the most similar. In stimulus C, most key design elements are placed along the cell borders. This can be attributed to causing the greatest apparent discrepancy in areas with highest heat.

To avoid the pitfalls of grid-based heatmaps potentially portraying distorted representations of gaze, more flexible alternatives can be seen as preferable. Gaussian pixel-based heatmaps with the size of the kernel increased can better represent the error of the gaze estimations. A larger kernel (see Fig. 6) reflects in the breadth of the area where each predicted fixation increases heat, avoiding the illusion of false pinpoint accuracy in favor of displaying the area where the gaze is likely to be found.

Until webcam-driven methods catch up to infrared gaze tracking, invention of new visualizations specifically for appearance-based gaze estimation can be encouraged. To accommodate the inherent estimation error of an appearance-based system, visualizations such as fuzzy heatmaps could infer the size of extracted areas of interest(AOIs) and the angular error to model the probability of gaze being focused on AOIs, or the binomial distributions of numbers of users seeing/not seeing particular AOIs.

Among the examined environmental factors (glasses, distance from the screen, webcam placement and illumination level), only the increase in distance failed to exert a significant effect on the proposed network to a greater magnitude than on infrared gaze tracking with the corneal reflection technique. This could be attributed to a 10 cm difference simply not being enough to impact the less accurate appearance-based model. Tobii's infrared tracking being affected by distance is within expectation, since this is addressed in its documentation.

Analysis of eye images from participants wearing glasses supports a suggestion that occlusion by glasses increases the on-screen distance error due to inferring with facial landmark recognition. Eyes with glasses are often off-center, some images further polluted by the reflection of light off the lenses. Normalization involving the removal of glasses from webcam footage may be fit to counter this effect.

Placement of the webcam in the environment directly interferes with appearance-based gaze estimation. If the webcam is placed below or on the side of the screen, the predicted gaze points shift in the direction opposite to the camera's placement (see Fig. 13). This transpositionary shift can be directly attributed to an expectation for the screen to be located below the camera. For example, with the webcam placed on the desk and gaze directed at the screen above it (an area that would be above the upper boundary of the screen, were the webcam display-mounted), the model clusters most estimated gaze points in the zone close to the upper boundary. It can be suggested that even with the webcam placed on top of the screen, a transpositionary shift could be observed if the camera were placed off-center, lower or higher. We infer that the use of an external webcam positioned centimeters higher than a built-in webcam of a laptop is what caused the shift of appearance-based gaze estimation downwards for all participants in experiments 2 and 3, as seen in Figs. 13, 7 and 6.

We suggest that new gaze estimation datasets featuring varied and annotated relative positions of the webcam and the screen, realistic illumination levels and head angles of the user gazing at a computer screen in naturalistic conditions, could be highly valuable for the development of more robust appearance-based models for gaze estimation in 2D screen coordinates. Models should be able to adapt to the screen-camera position automatically, since requiring the user to perform explicit actions to calibrate the relative screen-camera position for the model would hinder its usability, and thus its democratization.

Examination of indoor light conditions demonstrates that darkness and harsh light both cause the gaze points estimated by the proposed network to scatter further (see Fig. 13), with darkness being overall less damaging to accuracy than harsh light. Light overexposure and contrast issues result in images that are less sharp and overlaid with pronounced shadows, as seen Fig. 9. Meanwhile, color discrepancies typical of images captured in the dark present less of an issue since inputs are processed in greyscale. Notably, were the infrared eye tracker calibrated separately each time that illumination changes, the contrast in performance between the methods can be reasonably presumed to amplify.

When looking for solutions to democratize gaze tracking in the wild, following implications can be drawn about the applicability of calibrationless appearance-based gaze estimation using the proposed network (and potentially, webcam-driven gaze estimation of similar levels of accuracy):

- Lower accuracy of the appearance-based approach is not a fatal obstacle for insightful visualization, but should be accounted for during the plotting and interpretation of gaze.
- Heatmaps of appearance-based gaze estimations can portray broader areas where gaze is likely to be situated, in place of the exact position.
- Grid heatmaps can be used to visualize appearance-based gaze estimation results, with their granularity adjusted to screen size and angular error of the estimation method. Their reliability is higher if areas of interest are located squarely within the grid cells, and lower if they are found on the grid's borders.
- Gaussian pixel-based heatmaps can be adapted to the inaccurate nature of appearance-based gaze estimation by plotting fixations with a corresponding larger heat radius.
- Understanding heatmaps that portray the probable rather than the exact position of gaze can be bridged by knowledge of the user task and the stimulus involved. Identifying the focus of gaze reliant on expertise of the interpreter may prove inadequate with diminishing size and increasing proximity of gaze targets.
- Of the observed environmental factors, placement of the webcam has the greatest and most distinct effect, transposing the estimated gaze in the opposite direction from the shift of the camera from its "ideal" position above the screen. If all gaze estimations for a specific user are apparent as being transposed in a unified direction, clustering at higher density at an edge or in the corner of the screen, webcam placement is a probable cause.
- If users are more likely to be recorded in the dark, while wearing glasses, in harsh light (whether inside or outside), the error of appearance-based gaze estimation can be expected to be above the benchmark error.
- To make appearance-based gaze estimation robust against varied webcam models and positioning in real world conditions (built-in, external, display-mounted, on the desk or on a tripod), novel training datasets, preprocessing approaches and network architectures should be developed. Synthesis or purification of data of environmental factors (Zhao et al., 2019), as Fu et al. (2020) demonstrated for illumination in outdoor eye images, may present an efficient avenue to address these inherent issues.
- Optionally, considering the known impact of factors of the environment, if the nature of collection of webcam data allows it, participants should be encouraged to (a) position the webcam on top of the screen, (b) ensure they are in a moderately-lit room and (c) remove their glasses, unless essential for reading. Distance from the screen may still be impactful for some methods, though its significance could not be proven at this moment in time.

7. Limitations and future work

The proposed RAGE-net architecture was evaluated and developed using the MPIIFaceGaze and MPIIGaze datasets. While commonly used in literature, with MPIIFaceGaze in particular containing a genuine variety of illumination conditions and head poses, for subjects with different characteristics (e.g., age, ethnicity), its accuracy may vary. Other datasets, such as the ETH-XGaze (Zhang et al., 2020) could be used for further analysis of the performance of our architecture. Pre-training with another dataset is promising to further the model's estimation accuracy (Cheng and Lu, 2022), though not to a degree that would counteract the discussed constraints and challenges of appearance-based methods, since related works exhibit similar accuracy with pre-training (Huang et al., 2023).

RAGE-net is evaluated for desktop computer environments to potentially leverage the user's home webcam as an appearance-based gaze estimation method in screen coordinates. As such, its performance in different environments, such as retail shelves or mobile devices, may differ and should be compared with models designed with this focus (Senarath et al., 2022). It can be extrapolated that for democratization of gaze tracking on mobile devices, research of novel datasets and gaze estimation methods that account for environmental factors can be even more crucial due to higher variance in use conditions, distances between the screen and the front-facing camera and small screen size. For illustration, a phone screen that is 15 cm tall (a large phone such as the iPhone 16 Pro), held at a distance of 40 cm from the eyes (a typical viewing distance), occupies 20.55° of the visual angle, making the gaze estimation error about one fifth of the screen height.

The assessment of the impact of the environment on appearance-based gaze estimation is conducted solely with the RAGE-net method. RAGE-net's good performance relative to previous methods lends credibility to the position that less accurate appearance-based methods would perform either identically or worse when deployed in the same conditions. To avoid overgeneralization, it should be stated that different appearance-based methods may be more or less robust against the impact of individual environmental factors.

The participants in both experiments were young, majorly male people of Slavic ethnicity. The training of the network was conducted on a more diverse sample represented in the MPIIFaceGaze dataset. Findings made about interpretability of gaze heatmaps and the impact of environmental factors on estimation accuracy are therefore tied to data from this specific demographic evaluated by a more generalistic network.

Harsh light as an environmental factor is represented in the experiment by a condition where the scene is illuminated ambiently by a high-power ceiling light fixture. The images of the face are illuminated strongly and evenly. In future research aiming to explore the impact on light on performance of gaze estimation, varied angles for casting directional light (and hard shadows) on the face are an additional factor up for consideration.

Findings about environmental factors may or may not be replicated in other demographics. We surmise that if differences were found in other demographics, they would majorly be oriented towards making appearance-based estimation more rather than less strained to be used effectively under varied environmental conditions (e.g., older users with facial feature textures that deviate from the training data; bone structures that could amplify the effect of illumination at varying lighting angles).

Feature-based gaze estimation via the commercial Tobii TX300 eye tracker failed to provide gaze points for 37.16% of observations when the distance of participants from the screen is 80 cm (65 cm is recommended in documentation). The apparent accuracy of the infrared gaze estimation is therefore artificially inflated by how it reports its results, making it inapplicable to gaze estimation in real conditions where the eye-screen distance may vary. A full and fair confrontation between appearance and feature-based gaze estimation based on distance may warrant corneal reflection technique that makes estimations for every observation, even if conditions for it are suboptimal.

Being a calibrationless approach, RAGE-net model's gaze grid heatmaps show the individual's identity as a key covariate of their accuracy, corroborating known benefits of calibration (Li et al., 2020). To avoid a calibration step, research of approaches eliminating the impact of identity and facial features may yield higher consistency (Cheng et al., 2022).

8. Conclusion

This paper proposes RAGE-net, a novel convolutional neural network for appearance-based gaze estimation without calibration, which performs comparably to previous attention mechanism and ResNet-based models in realistic conditions (4.08° angular error with MPIIFaceGaze) while also being more computationally efficient. This is accomplished by reducing the network's inputs to eye images, passed via an input normalization pipeline into a ResNet18-inspired network, comprising the feature and attention branches and incorporating batch normalization and weight sharing between the left and right eye. Applicability assessment reveals that to prevent potential misinterpretation in gaze estimation conducted with a webcam, visualizations should be judiciously adapted to be commensurate with attainable levels of accuracy. For instance, opting for somewhat coarser-grained grid heatmaps–which mitigate disparity from infrared gaze tracking results–may be prudent for gaze tracking tasks where medium accuracy is sufficient. Effects of real-world conditions such as webcam positioning, occlusion by glasses and scene illumination can further exacerbate deviations in resulting estimates, justifying further research to pave the way for true democratization of gaze tracking.

CRediT authorship contribution statement

Eduard Kuric: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Peter Demcak:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing. **Jozef Majzel:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Giang Nguyen:** Conceptualization, Formal analysis, Methodology, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset and source code are available in a paper repository at: https://github.com/ragenetresearch/democratizing-eye-tracking-rage-net.

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